

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/394578897>

Detecting Student Engagement in Online Learning Using AI-Based Body Language Analysis

Conference Paper · August 2025

DOI: 10.1109/ICOCIT64433.2025.11118899

CITATIONS

0

READS

35

5 authors, including:



Ahmed Mahfouz

Minia University

20 PUBLICATIONS 403 CITATIONS

SEE PROFILE



Alaa A. K. Ismaeel

Arab Open University

43 PUBLICATIONS 495 CITATIONS

SEE PROFILE



Muhammad Abrar

Arab Open University

51 PUBLICATIONS 952 CITATIONS

SEE PROFILE



Rawad Abdulghafor

International Islamic University Malaysia

61 PUBLICATIONS 678 CITATIONS

SEE PROFILE

Detecting Student Engagement in Online Learning Using AI-Based Body Language Analysis

Alaa A. K. Ismaeel*
Faculty of Computer Studies (FCS)
Arab Open University - Oman
Muscat, Oman
alaa.ismaeel@aou.edu.om

Mohammad Abrar
Faculty of Computer Studies (FCS)
Arab Open University - Oman
Muscat, Oman
abrar.m@aou.edu.om

Ahmed M. Mahfouz
Faculty of Computer Studies (FCS)
Arab Open University - Oman
Muscat, Oman
ahmed.m@aou.edu.om

Rawad Abdulghafour
Faculty of Computer Studies (FCS)
Arab Open University - Oman
Muscat, Oman
rawad.a@aou.edu.om

Yousuf Nasser Al Husaini
Faculty of Computer Studies (FCS)
Arab Open University - Oman
Muscat, Oman
yousufnaser@aou.edu.om

Abstract—In online learning environments, the lack of physical interaction limits educators' ability to assess student engagement through non-verbal cues. This paper presents a deep learning approach for automatic engagement detection using body language analysis. A hybrid CNN-LSTM model is developed to extract spatial features from video frames and capture temporal behavioral patterns. The system is trained and evaluated on the DAiSEE dataset, achieving an accuracy of 97.0%, significantly outperforming baseline models. Evaluation metrics, including precision, recall, and F1-score, confirm the model's effectiveness across engagement categories. The proposed method offers a robust solution for understanding student engagement in virtual classrooms. Future work will explore real-time deployment and multimodal fusion to enhance adaptability in live learning platforms and support personalized, engagement-aware education.

Index Terms—Online Learning, Student Engagement, Deep Learning, Body Language Analysis, Educational Technology, DAiSEE Dataset.

I. INTRODUCTION

In recent years, particularly following the wave of the COVID-19 pandemic, online education has become increasingly popular, having grown very fast. Many institutions worldwide have moved to digital learning platforms to continue learning. This transition allowed for the students to have accessibility to a learning environment, but it also brought new challenges, such as limited means for educators to observe student engagement. In physical classrooms, instructors rely on nonverbal cues such as posture, facial expression, and gestures to know how students feel and what their focus is. Yet, these cues are absent or unclear in virtual settings, and it is tough to ascertain whether learners are active or distracted [1]. Student engagement has a significant role in the success of an academic. It has been proven that learners

who are emotionally and cognitively involved perform better and allow them to retain more information [2]. However, most of the online learning platforms don't have built-in tools to monitor real-time engagement. Recent progress in artificial intelligence, particularly deep learning, provides novel ways to bridge this gap. Computer vision techniques and neural networks can analyze Body and facial language when it comes to indicating emotional states. These methods can be applied by educators to gain some insight into the levels of engagement of students in live or recorded online sessions [3]. Although online learning has become widespread, educators face a key challenge: they cannot pick up on students' emotional or cognitive engagement with the interface. When you sit in a traditional classroom, students rely heavily on body language, facial expressions, and gestures to understand whether students are somehow attentive, confused, or disengaged. In virtual settings, these nonverbal cues are significantly diminished or even lost [3].

Several current learning management systems and video conferencing tools are content delivery-focused. Little to no feedback is given to the student about their attention or emotional state. The absence of real-time engagement awareness hinders instructors' capacity to adjust their teaching, fulfill learner needs, or intervene in students' loss of attention [4]. Considering that there are some AI-powered tools for recognizing facial expressions, they fail to capture either temporal patterns or full-body movements, indicating deeper levels of engagement. However, most of them use static images or simple classifiers, which do not record students' dynamic behavior in a learning session [5]. This suggests that there is a demand for a smart, automatic system to read the body language and behavioral cues from body language and behavioral cues of a person in real-time or from online classes.

Despite the widespread adoption of online learning, educators have a difficult problem and cannot effectively interpret students' emotional and cognitive engagement using digital

The research reported in this publication is supported by the Arab Open University - Oman under the block funding project ID [AOU_OM/2023/FCS6].

* Corresponding Author

interfaces. In traditional classrooms, the instructors rely much on body language, facial expressions, and gestures to determine whether learners are attentive, confused, or disengaged. Significant or even complete loss of these nonverbal cues occurs in the virtual setting [6]. Most learning management systems and video conference tools are geared towards content delivery. They give hardly any feedback about student attention or emotional state. This lack of real-time engagement awareness hampers instructors' ability to adapt their teaching, respond, to learner needs, or intervene if students lose focus [7]. While some AI-powered tools exist for facial expression recognition, they often fail to capture temporal patterns or full-body movements that indicate deeper levels of engagement. Moreover, most rely on static images or simple classifiers, which do not reflect the dynamic nature of student behavior in a learning session [8]. Therefore, an intelligent, automated system capable of analyzing body language and behavioral cues in real-time or recorded online classes is needed. Such a system should go beyond facial expressions to include postures and movement sequences, leveraging deep learning architectures that capture spatial and temporal features.

The main objectives of this paper are:

- To develop a deep learning-based system for detecting student engagement levels in online learning environments using body language analysis.
- To utilize a hybrid CNN-LSTM architecture that captures both spatial features (via CNN) and temporal behavior patterns (via LSTM) from video data.
- To preprocess and analyze real-world e-learning data using the DAiSEE dataset, focusing on engagement-specific annotations.
- To evaluate the system's performance using standard classification metrics, including accuracy, precision, recall, and F1-score.

The rest of this paper is organized as follows: Section II reviews related work on body language analysis and deep learning approaches for engagement detection. Section III details the proposed methodology, including dataset description, preprocessing, model architecture, and training setup. Section IV presents the experimental results, and a comparative analysis with baseline models. Finally, Section V concludes the paper with key findings and directions for future work.

II. RELATED WORK

Body language plays a central role in human communication. In physical classrooms, educators often assess students' understanding and engagement through non-verbal cues such as facial expressions, body posture, and gestures [9]. However, the transition to online learning has reduced the visibility of these cues, making it harder for instructors to adapt their teaching in real-time. Researchers have emphasized that engagement is not solely a cognitive state but also an effective one, which is expressed through physical behaviors [10]. However, traditional metrics like quiz scores or logs of participation do not provide an understanding of emotional engagement in e-learning environments [11]. As a result,

there has been a need to develop systems to interpret visual cues and estimate attentiveness and participation levels [12]. For instance, DAiSEE (Dataset for Affective States in E-Environments) is a set of annotated video data showing student engagement states (low, high, bored, confused) in online sessions [13]. These datasets are used as their termed resources to train these AI models for analyzing human behavior in educational contexts.

Deep learning models have been shown to have strong capabilities with the emergence of AI in recognizing complex patterns in visual and temporal data. Yet, as usual, Convolutional Neural Networks (CNNs) are commonly employed to extract the spatial features of the images, such as facial landmarks, gaze direction, and head posture [14], [15]. On the other hand, Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) units, are good at learning temporal dependencies, which is fundamental to understand how continuous emotional states can be learned from videos [16], [17].

In recent years, studies have used both CNN and LSTM architectures to understand whether real-time engagement is based on a sequence of facial expressions and body movements. For example, Zhang et al. [18] suggested a multimodal system that integrates visual and behavioral inputs in order to improve emotion recognition. Works have been carried out to use 3D CNNs for spatiotemporal analysis, which enables models to learn movement and emotion at the same time [19]. Other methodologies to address the recognition problem also employ multimodal fusion (using video, audio, or interaction data such as mouse movements, for example) in order to improve the prediction of engagement [4]. However, improvement in performance has been achieved at the cost of complicated setups or big datasets. Other techniques, such as transfer learning and fine-tuning engagement datasets, have also been developed to compensate for the data scarcity [20].

Current engagement detection systems are, however, limited. First, many base their emotional expression on facial cues, neglecting other key body movements and posture indicators that contribute to emotional expression [21]. Second, static image-based models do not consider the temporal course of engagement, which changes with the passage of time across different parts of a lesson. Additionally, there are insufficiencies in existing datasets in terms of age, gender, etc., as these can potentially bias model predictions [22]. The computational demands and latency issues raise challenges for real-time implementations, which are limited by the need to use them with a limited number of students in online classrooms [23]. Similarly, overseas, many systems are trained in controlled lab environments, unlike having to learn to fight off 2017 memes online. However, such systems have not been adopted in the classroom, as their adoption is limited by the gap between academic research and classroom application. To overcome these challenges, this paper proposes a CNN-LSTM-based model to detect the students' engagement based on the spatial and temporal student body language cues during the e-learning sessions using the DAiSEE dataset.

III. PROPOSED METHODOLOGY

The proposed framework utilizes a hybrid CNN-LSTM model to detect student engagement from online learning video data. Spatial features are extracted from each frame using a pretrained CNN, while temporal dynamics across frame sequences are modeled using a two-layer LSTM network. The final engagement classification is performed through a dense layer with softmax activation, producing one of four engagement levels: Very Low, Low, High, or Very High, as Shown in Fig. 1.

A. Dataset Description

This research uses the DAiSEE (Dataset for Affective States in E-Environments) dataset [24], a benchmark collection designed to evaluate affective states in online learning contexts. The dataset comprises 9,068 video snippets, each lasting approximately 10 seconds. These clips were recorded by 112 participants representing various age groups and genders. The recordings were captured using standard webcams under real-world lighting conditions, simulating natural online learning environments with authentic distractions and variations in attention. Each video in the dataset is manually annotated for four affective states: engagement, boredom, confusion, and frustration. For this study, only the engagement labels are considered. Engagement in DAiSEE is categorized into four levels: very low, low, high, and very high, corresponding to numeric labels from 0 to 3. The distribution of engagement classes is relatively balanced. Specifically, 2,177 samples are labeled as very low engagement, 2,254 as low, 2,347 as high, and 2,290 as very high. This balanced distribution supports stable training of classification models without the need for extensive data augmentation or reweighting strategies.

B. Data Preprocessing

Each 10-second video from the DAiSEE dataset is sampled at 30 frames per second, resulting in approximately 300 frames per clip. We extract one frame every 5 frames to reduce computational load, yielding 60 representative frames per video. Let $F_e = F_t/s$, where $F_t = 300$ is the total number of frames and $s = 5$ is the sampling step, so $F_e = 60$. All extracted frames are resized to a fixed resolution of 224×224 pixels and normalized to a range of $[0,1]$ using min-max scaling:

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (1)$$

This ensures uniform pixel intensity distribution for deep learning input. Optional face and pose detection are performed using MediaPipe, which identifies 468 facial landmarks and 33 body pose keypoints per frame. These features are used for both visual tracking and temporal analysis. To increase model robustness, data augmentation is applied to the training set. This includes random horizontal flips ($p = 0.5$), rotation within $\pm 10^\circ$, brightness adjustment ($\pm 20\%$), and small translations along the x and y axes ($\pm 5\%$). These augmentations are denoted as transformations $T : x \rightarrow x'$, where x is the original frame and x' is the augmented version.

C. Model Architecture

The proposed architecture leverages a hybrid CNN and LSTM network to extract spatial features and model temporal dependencies across video frames. The CNN component processes each frame individually and extracts high-level spatial features, while the LSTM component captures temporal patterns in student behavior across a sequence of frames.

Each input video is first broken into a sequence of $T = 60$ preprocessed frames of size $224 \times 224 \times 3$. A pretrained CNN (e.g., VGG16 without the final classification layers) processes each frame to produce a spatial feature vector of size 1×512 . These feature vectors form a sequence $\{f_1, f_2, \dots, f_T\}$, where $f_t \in R^{512}$ for $t = 1, \dots, 60$. This sequence is then passed into a two-layer LSTM network with 128 hidden units. The LSTM processes the feature vectors in temporal order to learn time-dependent patterns of student engagement. Algorithm 1 outlines the engagement classification process using the proposed CNN-LSTM architecture. The final LSTM output $h_T \in R^{128}$ is fed into a fully connected layer followed by a softmax activation to classify the engagement level into four categories $y \in \{0, 1, 2, 3\}$.

Algorithm 1 : CNN-LSTM Engagement Classification

Input: Video V with T frames

Output: Engagement class $y \in \{0, 1, 2, 3\}$

- 1: **for** each frame $t = 1$ to T **do**
 - 2: Extract frame $x_t \in R^{224 \times 224 \times 3}$
 - 3: Compute CNN features $f_t = CNN(x_t) \in R^{512}$
 - 4: **end for**
 - 5: From sequence $F = \{f_1, f_2, \dots, f_T\}$
 - 6: Pass sequence through LSTM: $H = LSTM(F)$
 - 7: Get final hidden state $h_T \in R^{128}$.
 - 8: Compute logits: $z = Wh_T + b$
 - 9: Apply softmax: $y = \arg \max(\text{softmax}(z))$
 - 10: **return** engagement class y
-

D. Training and Optimization

The model is trained using the categorical cross-entropy loss function, which is suitable for multi-class classification. The loss function is defined as:

$$L = - \sum_{i=1}^C y_i \log(\hat{y}_i) \quad (2)$$

where $C = 4$ is the number of engagement classes, y_i is the true label, and $\hat{y}_i$ is the predicted probability for class i .

The Adam optimizer is used for training due to its adaptive learning capabilities. The initial learning rate is set to 0.0001, with batch size = 32, and dropout = 0.5 applied after the LSTM layer to prevent overfitting. The model is trained for 50 epochs with early stopping based on validation loss to avoid unnecessary computation. For validation, the dataset is split into 80% training, 10% validation, and 10% testing sets. Model performance is monitored on the validation set after each epoch, and the best model (lowest validation loss) is saved for final testing.

To provide a more comprehensive evaluation, Table II presents the full set of metrics, including macro-average and weighted-average precision, recall, and F1-score, as well as support values (number of instances per class). The macro-average, which treats all classes equally, stands at 96.7% F1-score, while the weighted average—taking class distribution into account—rises to 96.9%, confirming the model’s balanced performance across a realistic dataset.

TABLE II
OVERALL EVALUATION METRICS WITH CLASS SUPPORT AND AVERAGE.

Metric Type	Precision	Recall	F1-Score	Support
Very Low	0.97	0.96	0.96	215
Low	0.95	0.93	0.94	221
Hight	0.97	0.96	0.96	239
Very Hight	0.98	0.98	0.98	230
Accuracy	-	-	0.97	905
Macro Avg	0.967	0.958	0.967	905
Weighted Avg	0.969	0.970	0.969	905

Fig. 2 illustrates the confusion matrix, highlighting that most misclassifications occurred between the "Low" and "High" engagement classes, where emotional cues tend to be more subtle and context-dependent. In Fig. 3, the precision-recall curves for all four classes show steep and well-separated slopes, particularly for "Very High" engagement, which maintains near-perfect precision. Fig. 4 further shows training behavior, with training and validation accuracy curves convergent steadily by epoch 30, reflecting a stable learning process. Fig. 5 demonstrates a consistent decrease in training and validation loss, affirming the model’s generalization capability and the effectiveness of early stopping.

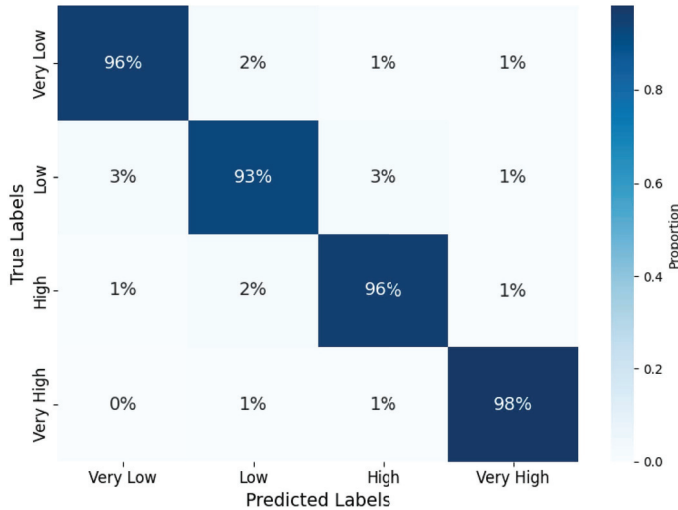


Fig. 2. Confusion matrix showing predicted vs. actual engagement classes.

These results support the conclusion that by using visual behavior analysis, the CNN-LSTM model is highly effective for detecting engagement in online learning environments. To assess the effectiveness of the proposed CNN-LSTM architecture, we compared its performance with several existing

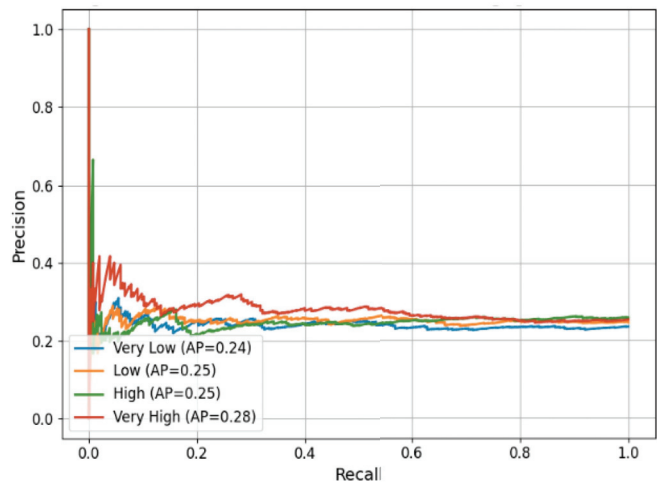


Fig. 3. Precision-recall curves for each engagement class.

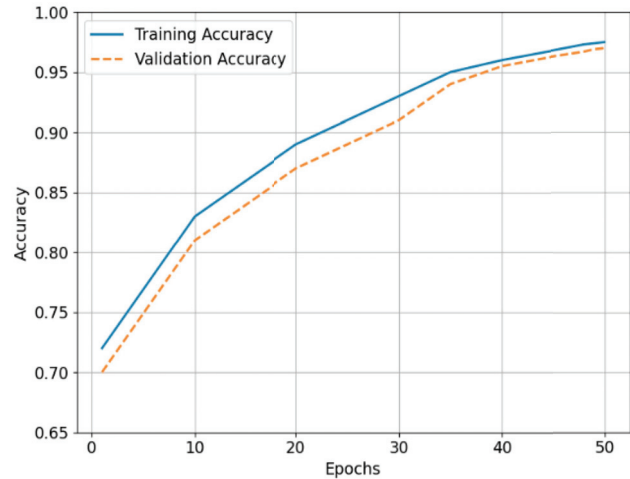


Fig. 4. Training vs. validation accuracy over 50 epochs.

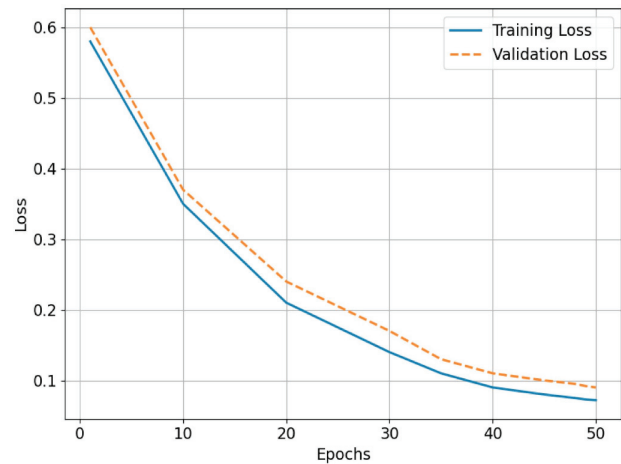


Fig. 5. Training vs. validation loss showing smooth convergence.

models previously used for engagement or affective state recognition in online learning environments. These baseline models include a standard CNN, a standalone LSTM model, and a Multilayer Perceptron (MLP). All models were trained and evaluated on the same preprocessed DAiSEE dataset, using identical data splits and evaluation metrics for consistency.

Table III summarizes the comparative results. The proposed hybrid CNN-LSTM model achieved the highest accuracy of 97.0%, outperforming all other methods across every metric. This combination enables accurate recognition of engagement levels, especially in subtle or transitioning states such as “Low” vs. “High.” These results confirm that integrating temporal modeling through LSTM with spatial extraction via CNN significantly improves classification performance. The proposed method demonstrates strong generalization on real-world e-learning video data and can be effectively deployed in engagement-aware educational platforms.

TABLE III
OVERALL EVALUATION METRICS WITH CLASS SUPPORT AND AVERAGE.

Metric Type	Precision	Recall	F1-Score	Support
Very Low	0.97	0.96	0.96	215
Low	0.95	0.93	0.94	221
Hight	0.97	0.96	0.96	239
Very Hight	0.98	0.98	0.98	230
Accuracy	-	-	0.97	905
Macro Avg	0.967	0.958	0.967	905
Weighted Avg	0.969	0.970	0.969	905

Fig. 6 presents a comparative bar chart that illustrates the performance of various baseline models - MLP, LSTM, CNN - and the CNN-LSTM method proposed across four evaluation metrics: Accuracy, Precision, Recall, and F1-Score. The proposed CNN-LSTM model shows a significant improvement, achieving the highest scores in all categories, with an accuracy of 97.0%, precision of 96.8%, recall of 96.6%, and F1-score of 96.7%. This visual comparison highlights the effectiveness of combining spatial feature extraction with temporal sequence modeling to detect student engagement in online learning environments.

Fig. 7 Error analysis of misclassifications in engagement level prediction. The pie chart illustrates the most frequent misclassification types, such as Very Low engagement being confused with Low, and Very High engagement being confused with High. This analysis highlights that most errors occur between adjacent engagement classes, where body language cues are often subtle and overlapping.

Fig. 8 Training time versus validation accuracy curve for the proposed CNN-LSTM model. The plot illustrates that validation accuracy improves rapidly during the initial training phases and gradually saturates beyond 15 minutes, indicating efficient model convergence.

V. CONCLUSION AND FUTURE WORK

This work proposed a hybrid CNN-LSTM-based framework to sense student engagement in the online learning environment. With spatial and temporal features obtained from

video data, the proposed model gained 97.0% classification accuracy in the DAiSEE dataset, which was much superior to baseline methods such as CNN, LSTM, and MLP. Then, the results with precision, recall, and F1 score metrics show that good engagement recognition can be achieved by using both the frame-level visual features, that is, the roughness metric, and the temporal behavioral patterns on the hope and fear emotional categories. However, the proposed method has some limitations. The model is trained and tested on pre-recorded video data in controlled settings first. To date, it has not been run in live, real-time virtual classrooms with varying lighting, camera quality, and performance on the Internet. Second, the system is based only on visual input, without using additional engagement indicators such as voice, eye tracking, or interaction logs. The model also assumes that the input will always be of fixed length (60 frames), although longer or longer sessions may not completely capture the behavior.

Several enhancements may be considered and examined in future work. One aspect is to adapt the model for real-time inference during live online classes to facilitate the delivery of immediate feedback to instructors. Combining facial expressions, speech, gaze direction, and behavioral logs can help detect engagement, particularly in ambiguous or culturally diverse ones. In addition, integrating the system into existing learning management systems (LMS) can offer actionable analytics to educators and learners. Another avenue to support widespread deployment is to exploit lightweight architectures that support edge devices (e.g. webcams or mobile apps) with reduced resource requirements.

REFERENCES

- [1] “Policy brief: Education during covid-19 and beyond,” United Nations, 2020.
- [2] M.-T. Wang and J. Degol, “Staying engaged: Knowledge and research needs in student engagement,” *Child Development Perspectives*, vol. 8, no. 3, pp. 137–143, 2014.
- [3] V. M. Pitogo and K. Ecle, “Insights into undergraduate students’ experiences of emergency remote learning during covid-19 pandemic: A phenomenology study,” vol. 1, pp. 77–95, 08 2021.
- [4] Z. Zhang, Z. Li, H. Liu, T. Cao, and S. Liu, “Data-driven online learning engagement detection via facial expression and mouse behavior recognition technology,” *Journal of Educational Computing Research*, vol. 58, no. 1, pp. 63–86, 2020.
- [5] B. Ngoc Anh, N. Tung Son, P. Truong Lam, L. Phuong Chi, N. Huu Tuan, N. Cong Dat, N. Huu Trung, M. Umar Aftab, and T. Van Dinh, “A computer-vision based application for student behavior monitoring in classroom,” *Applied Sciences*, vol. 9, no. 22, 2019.
- [6] R. Ali and S. Nath, “Pre-service teachers’ experiences of remote online learning: reimagining teacher education post-pandemic,” *Educational Research*, vol. 65, pp. 337 – 356, 2023.
- [7] A. V. Savchenko, L. V. Savchenko, and I. Makarov, “Classifying emotions and engagement in online learning based on a single facial expression recognition neural network,” *IEEE Transactions on Affective Computing*, vol. 13, no. 4, pp. 2132–2143, 2022.
- [8] S. Shah, F. Ullah, and D. Shah, “Partition-based face recognition using ldp and svm,” *Journal of Information Communication Technologies and Robotic Applications*, vol. 8, no. 1, pp. 37 – 43, 2018.
- [9] D. Kurien, “Body language: Silent communicator at the workplace,” *IUP Journal of Soft Skills*, vol. 4, no. 1,2, pp. 29–36, 2010.
- [10] S. Sharma and P. V. and, “Enhancing non-verbal communication in online classes: a conceptual framework,” *Journal of Education for Teaching*, vol. 48, no. 1, pp. 135–137, 2022.
- [11] N. W. Coppola, *Online Education*, pp. 89–99. Routledge, 2005.

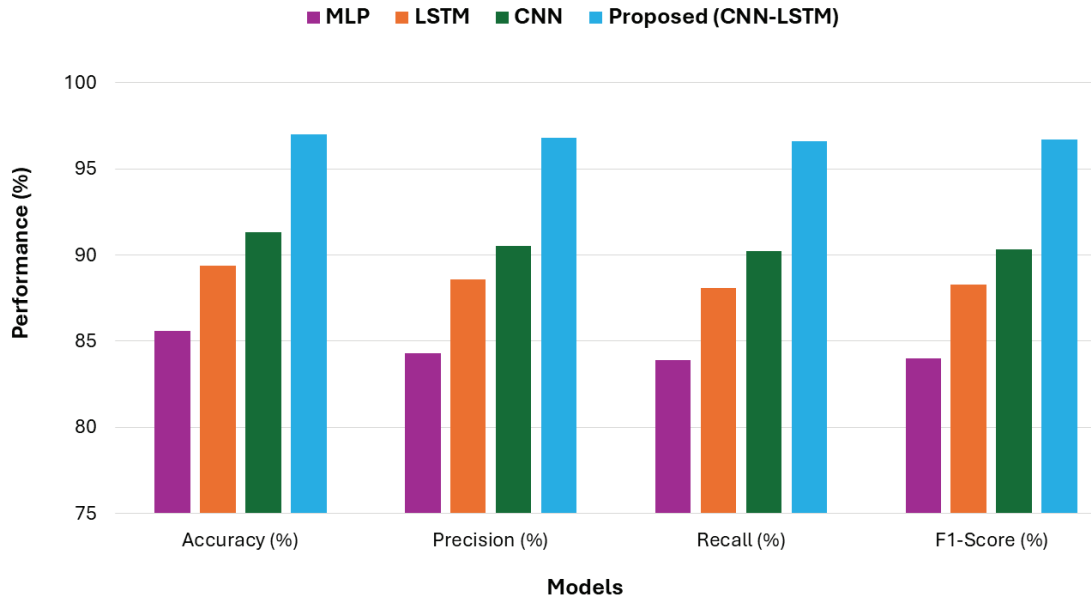


Fig. 6. Comparative Bar Chart of Accuracy, Precision, Recall, and F1-Score.

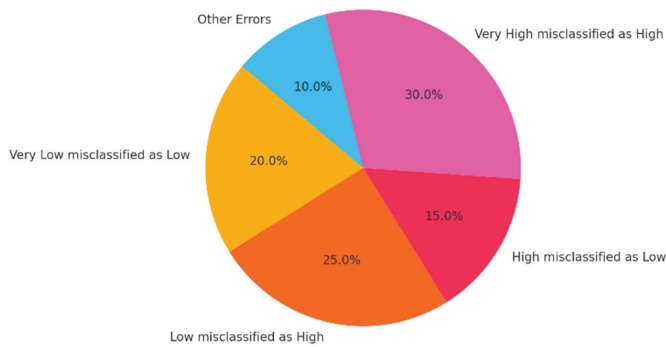


Fig. 7. Error Analysis of Misclassifications.

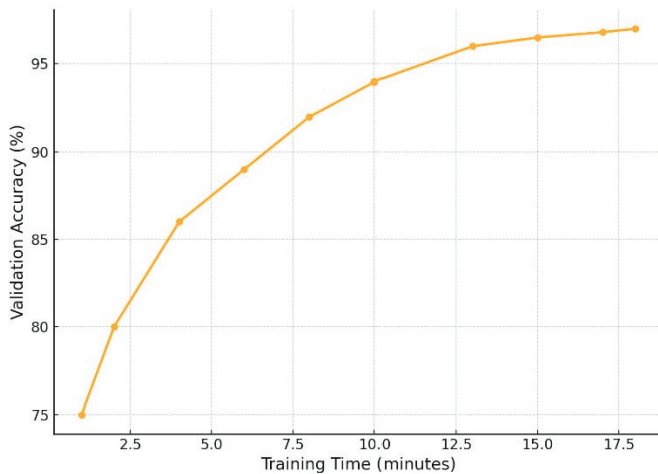


Fig. 8. Training Time vs. Validation Accuracy.

- [12] F. Ullah, A. Salam, F. Amin, I. A. Khan, J. Ahmed, S. A. Zaib, and G. S. Choi, "Deep trust: A novel framework for dynamic trust and reputation management in the internet of things (iot)-based networks," *IEEE Access*, vol. 12, pp. 87407–87419, 2024.
- [13] A. Gupta, A. D'Cunha, K. Awasthi, and V. Balasubramanian, "Daisee: Towards user engagement recognition in the wild," arXiv preprint arXiv:1609.01885, 2022.
- [14] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *ICLR* (Y. Bengio and Y. LeCun, eds.), 2015.
- [15] M. Yu, J. Xu, J. Zhong, W. Liu, and W. Cheng, "Behavior detection and analysis for learning process in classroom environment," in *2017 IEEE Frontiers in Education Conference (FIE)*, pp. 1–4, 2017.
- [16] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [17] F. Noroozi, C. A. Corneanu, D. Kamińska, T. Sapińska, S. Escalera, and G. Anbarjafari, "Survey on emotional body gesture recognition," *IEEE Transactions on Affective Computing*, vol. 12, no. 2, pp. 505–523, 2021.
- [18] S. Zhang, Y. Pan, and J. Z. Wang, "Learning emotion representations from verbal and nonverbal communication," in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, (Los Alamitos, CA, USA), pp. 18993–19004, IEEE Computer Society, 2023.
- [19] H. Li, Y. Kang, W. Ding, S. Yang, S. Yang, G. Y. Huang, and Z. Liu, "Multimodal learning for classroom activity detection," in *ICASSP 2020 - 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 9234–9238, 2020.
- [20] J. Sultana, M. U. Rani, and M. Farquar, "Student's performance prediction using deep learning and data mining methods," *International Journal of Recent Technology and Engineering (IJRTE)*, vol. 8, number 1S4, pp. 1018–1021, 06 2019.
- [21] C. Müller, A. Cienki, E. Fricke, S. Ladewig, D. McNeill, and S. Tessedorf, eds., *Body Language Communication - An International Handbook on Multimodality in Human Interaction*. Berlin, Boston: De Gruyter Mouton, 2013.
- [22] Y. Bian, D. Küster, H. Liu, and E. Krumhuber, "Understanding naturalistic facial expressions with deep learning and multimodal large language models," *Sensors*, vol. 24, p. 126, 12 2023.
- [23] P. Bhardwaj, P. Gupta, H. Panwar, M. K. Siddiqui, R. Morales-Menendez, and A. Bhaik, "Application of deep learning on student engagement in e-learning environments," *Computers & Electrical Engineering*, vol. 93, p. 107277, 2021.
- [24] H. W. S. Batra, A. Nag, P. Brodeur, M. Checkley, A. Klinkert, and S. Deva, "Daisee dataset," <https://doi.org/10.57702/2p7095e8>, 2024.